

Summary of Revisions from Version 1 to Version 2

PhysioRisk: an interpretable survival-modeling framework for separating demographic and physiological mortality risk

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Version 2 retains the original study question, NHANES 1999–2016 data source, all-cause mortality endpoint, two-axis PhysioRisk construction, and temporal development/validation split. It substantially revises the composite risk score, corrects one effect-size estimate, expands validation diagnostics and uncertainty quantification, and clarifies model provenance and interpretation. The principal changes are summarized below.

1. Composite risk score reconstructed

Version 1 used a unit-weight TotalRisk formed by adding DemoRisk to standardized PhysioRisk. Reconstruction showed that standardizing PhysioRisk changes its coefficient scale, so the unit-weight sum is not generally equivalent to the joint Cox linear predictor. Version 2 therefore uses CalibratedTotalRisk as the primary composite: $1.0586578128 \times \text{DemoRisk} + 0.3563514639 \times \text{PhysioRisk}$. Both coefficients were estimated jointly using only the 1999–2010 development cohort and then fixed before 2011–2016 temporal validation. The historical unit-weight TotalRisk is retained only for provenance/supplementary comparison.

2. PhysioRisk hazard-ratio estimate corrected

The Version 1 HR of approximately 2.77 per SD for PhysioRisk came from an unpenalized Cox evaluation fit that did not converge. Version 2 uses a convergence-controlled Cox fit and reports HR 1.387 (95% CI 1.338–1.439) per temporal-validation-cohort SD. PhysioRisk discrimination and age correlation remain C-index 0.654 (95% CI 0.624–0.685) and $r=0.106$, respectively. The correction changes the magnitude of the reported hazard gradient but not the conclusion that age- and sex-adjusted physiological variation retains mortality information.

3. Uncertainty and direct model comparisons expanded

Version 2 adds participant-level bootstrap uncertainty (2,000 replicates) for C-indices and paired confidence intervals for prespecified discrimination contrasts. CalibratedTotalRisk improved C-index over DemoRisk by $\Delta C=0.01628$ (95% CI 0.01109–0.02199). Its difference from PhenoAge NCP was $\Delta C=0.00205$ (95% CI –0.00326 to 0.00748), providing no evidence of a discrimination difference. The PhysioRisk versus PhenoAge NCP acceleration contrast favored PhysioRisk numerically but remained inconclusive because its paired interval included zero.

4. Calibration, censoring robustness, and proportional-hazards diagnostics added

Version 1 emphasized relative risk ranking and did not present a full absolute-risk calibration analysis. Version 2 adds 5- and 8-year fixed-horizon calibration, observed-to-expected ratios, horizon-specific calibration slopes and intercepts, IPCW Brier scores, Uno C, cumulative/dynamic AUC, reverse Kaplan–Meier follow-up, and numbers at risk. Mean predicted and observed mortality were closely aligned at 5 years (4.610% vs 4.653%) and 8 years (8.571% vs 8.826%). Version 2 also reports that CalibratedTotalRisk violates strict proportional hazards and that its association strengthened over follow-up; this limits interpretation of a single constant hazard ratio.

5. Cohort flow, missingness, and selection effects made explicit

The revised manuscript reconstructs the full cohort flow: 92,062 source-cycle participants → 53,255 mortality-eligible participants → 19,829 biomarker-complete participants → 13,090 development and 6,739 temporal validation participants. It reports the 33,426 mortality-eligible participants excluded for missing required fields, notes their higher observed death proportion, and states clearly that analyses are unweighted complete-case model evaluations rather than population-representative NHANES estimates. The smaller 8-year risk-set tail is also reported explicitly.

6. Model-development provenance clarified

Version 1 stated broadly that preprocessing and model construction were estimated in the training cohort and then applied unchanged to temporal validation. Subsequent provenance reconstruction supports the preserved final scoring and evaluation path but cannot establish that every earlier decision leading to selection of the final two physiological axes was entirely development-only. Version 2 states this limitation directly. It distinguishes that historical uncertainty from the CalibratedTotalRisk recombination coefficients, which were demonstrably estimated in development only and frozen before temporal evaluation.

7. Interpretation and comparator claims narrowed

Version 2 reframes the contribution as demographic–physiological risk decomposition rather than predictive superiority. PhenoAge NCP is explicitly described as a no-CRP proxy rather than the full published PhenoAge implementation. The revised text distinguishes total mortality prediction from age-adjusted physiological deviation, clarifies that DemoRisk is not the Cox baseline hazard, and explains why residualizing biomarkers before risk modeling is not mathematically equivalent to residualizing an aggregate biological-age score afterward. PhysioRisk is presented as a statistical physiological-risk component, not as a definitive biological-age measure or causal estimate of aging.

8. Validation status and limitations strengthened

Version 2 remains an internal temporal validation within NHANES; no independent external cohort has been added. The revised manuscript states this more prominently and avoids claims of transportability or clinical utility. It also adds or expands limitations concerning complete-case selection, the no-CRP comparator, reconstructed model-selection history, fixed-horizon calibration uncertainty, proportional-hazards violation, lack of subgroup/decision-utility analysis, and the limited precision of the 8-year tail.

Overall interpretation

The revised results continue to support the central premise that mortality risk can be represented as separable demographic and physiological components. DemoRisk accounts for most overall ranking, while PhysioRisk remains weakly correlated with chronological age and contributes incremental prognostic information when recombined with demographic risk. Version 2 does not support a claim that the framework is superior to PhenoAge NCP; its principal contribution is the explicit and interpretable decomposition of the two sources of risk. Independent external validation remains the principal next step.